

Progress Report

- Increment 1 -

Group #1

Please use this template to describe your progress on the group project in the latest increment. Please do not change the font, font size, margins or line spacing. All the text in italic should be removed from your final submission.

1) Team Members

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2) Project Title and Description

Shiftsync

An employee scheduler application that allows managers to automate scheduling processes for their business. Users can log in as an employee or admin/manager. Employees can add their schedule and request to swap shifts. A manager/admin can add/remove employees, generate a master schedule accommodating to everyone's availability. Both can have a calendar view of their schedule.

3) Accomplishments and overall project status during this increment

During this increment, we have successfully finalized the foundational elements of the project. The structure of the Flask application and the SQLite database has been completely established and linked. We have implemented the authentication system, which includes the functionalities for user login and logout, as well as secure password hashing. The pages for creating new users and logging in existing users are fully operational and integrated with session management. The layouts for the employee dashboard and availability page have been developed. At this point, the core framework of the system is complete, and the project is in accordance with the initial scope outline for Increment 1. While the full scheduling automation has not yet been implemented, the system infrastructure required to support it is now fully in place. We are now ready to proceed with the implementation of scheduling logic and rule validation in the upcoming increment.

4) Challenges, changes in the plan and scope of the project and things that went wrong during this increment

One of the main challenges during this increment was getting the integration right between components created by different team members. Since the authentication system, password pages, and dashboard were developed separately, we needed to ensure that the database connections and session handling were consistent across all modules. We were able to overcome this by keeping up regular communication and testing after merging branches. Another shift in our plan was to start implementing password hashing sooner than we had initially talked about. Although we originally intended to concentrate solely on basic login gestures, we opted to add hashed password storage now to boost security and prevent the need for refactoring down the line. All in all, we didn't run into any major problems, but we did have to do some minor debugging to make sure

database queries, session validation, and page routing were all working correctly. We were able to handle these through testing and making iterative fixes.

Team Member Contribution for this increment

Jaliah

- a) Progress Report:
Wrote the section describing project setup, database integration, and authentication system development.
- b) Requirements and Design Document:
Defined system architecture, database schema for EmployeeInfo, and authentication workflow design.
- c) Implementation and Testing Document:
Documented database initialization steps, authentication route testing, and login/logout validation cases.
- d) Source Code:
 - Created and configured the SQLite database (ShiftSyncDB.db)
 - Developed init_dp.py to initialize the database schema
 - Implemented insertinto.py for inserting initial employee records
 - Established database connection logic in main.py
 - Implemented login/logout routes and session management in main.py
- e) Video

Nikki did the video.

Annika

- f) Progress Report:
Wrote the section describing dashboard development and user interface progress.
- g) Requirements and Design Document:
Defined dashboard layout structure and outlined future scheduling display requirements.
- h) Implementation and Testing Document:
Documented dashboard route logic in main.py
- i) Source Code:
 - Developed templates/dashboard.html
 - Implemented dashboard route logic in main.py
 - Integrated session checks to ensure only authenticated users can access the dashboard
- j) Video

Nikki did the video.

Isabella

- k) Progress Report:
Contributed to the section describing password functionality and security improvements.
- l) Requirements and Design Document:
Helped define password creation requirements and authentication validation rules.
- m) Implementation and Testing Document:
Documented password hashing implementation, login validation, and test cases for successful and failed authentication.
- n) Source Code:

- Implemented password handling logic inside main.py
- Developed templates/newuser.html for new user password creation
- Developed templates/existinguser.html for returning user login
- Integrated secure password hashing before storing credentials in ShiftSyncDB.db
- Ensured consistent session validation after login

o) Video

Nikki did the video.

Nikki

p) Progress Report:

Contributed to the section outlining availability feature development.

q) Requirements and Design Document:

Defined availability input requirements and constraints for future scheduling logic.

r) Implementation and Testing Document:

Document from validation and input handling for availability selection.

s) Source Code:

- Developed the availability input template (availability page inside templates/)
- Implemented availability routing logic in main.py
- Integrated basic validation for day and time selection

t) Video

Nikki did the video.

5) Plans for the next increment

Implement scheduling and validation rules, store availability and shifts, generate/display weekly schedules, and show clear error messages when rules are violated (ex-too many employees scheduled at once, max shifts per week, max hours per week)

6) Stakeholder Communication

Dear Stakeholders,

We are delighted to announce the successful completion of the initial development phase of ShiftSync. Throughout this phase, our team concentrated on establishing the essential components necessary for secure user access and future scheduling functionalities.

We have implemented a fully functional login system that allows employees to securely access the platform. New users can create credentials, and returning users can log in and out of their accounts. To protect sensitive information, passwords are securely stored using industry-standard hashing practices.

Furthermore, the core application structure and database foundation have been set up. This guarantees that employee information and future scheduling data can be stored, retrieved, and managed effectively. The layout for the employee dashboard has also been created, along with an interface for updating availability that will facilitate scheduling features in the subsequent phase.

Although this phase primarily concentrated on infrastructure, we performed comprehensive integration testing to ensure that all components function together reliably. Minor modifications were made during development

to enhance security practices earlier than originally planned, thereby minimizing the need for rework in the future.

At this stage, the project is progressing according to scope and expectations. The system's foundation is stable, secure, and ready for the next development phase, which will focus on scheduling logic, rule enforcement (such as maximum shifts and hours), and automated validation messaging.

We look forward to sharing continued progress as we expand the platform's scheduling capabilities.

Sincerely,
The ShiftSync Team

7) Link to video

<https://drive.google.com/file/d/1P7ygdrI3IZ7E062ch3yli4ZmfOvCqbx/view?usp=sharing>